Database Tables

1) Users

```
CREATE TABLE Users (
    user_id SERIAL PRIMARY KEY,
    user_name VARCHAR(100) NOT NULL,
    user_email VARCHAR(100) UNIQUE NOT NULL,
    user_phone VARCHAR(20),
    user_password VARCHAR(255) NOT NULL,
    user_role VARCHAR(20) DEFAULT 'customer' CHECK (user_role IN ('customer', 'admin'))
);
```

2) Hotel

```
CREATE TABLE Hotel (
    hotel_id SERIAL PRIMARY KEY,
    hotel_name VARCHAR(100) NOT NULL,
    hotel_location VARCHAR(100) NOT NULL,
    hotel_rating INT CHECK (hotel_rating BETWEEN 1 AND 5),
    hotel_description TEXT
);
```

3) RoomType

```
CREATE TABLE RoomType (
    room_type_id SERIAL PRIMARY KEY,
    hotel_id INT REFERENCES Hotel(hotel_id) ON DELETE CASCADE,
    room_type_name VARCHAR(50) NOT NULL,
    price_per_night INT NOT NULL,
    total_capacity INT NOT NULL CHECK (total_capacity >= 0)
);
```

4) Room

```
CREATE TABLE Room (
    room_id SERIAL PRIMARY KEY,
    room_type_id INT REFERENCES RoomType(room_type_id) ON DELETE CASCADE,
    room_number VARCHAR(10) NOT NULL,
    is_available BOOLEAN DEFAULT TRUE,
    UNIQUE(room_type_id, room_number)
);
```

5) Airline

```
CREATE TABLE Airline (
    airline_id SERIAL PRIMARY KEY,
    airline_name VARCHAR(100) NOT NULL
);
```

6) Flight

```
CREATE TABLE Flight (
    flight_id SERIAL PRIMARY KEY,
    airline_id INT REFERENCES Airline(airline_id) ON DELETE CASCADE,
    departure_city VARCHAR(100) NOT NULL,
    arrival_city VARCHAR(100) NOT NULL,
    departure_time TIMESTAMP NOT NULL,
    arrival_time TIMESTAMP NOT NULL,
    flight_price INT NOT NULL,
    available_seats INT NOT NULL CHECK (available_seats >= 0)
);
```

7) HotelBooking

```
CREATE TABLE HotelBooking (
    hotel_booking_id SERIAL PRIMARY KEY,
    user_id INT REFERENCES Users(user_id) ON DELETE CASCADE,
    room_id INT REFERENCES Room(room_id) ON DELETE CASCADE,
    check_in DATE NOT NULL,
    check_out DATE NOT NULL,
    total_cost INT NOT NULL,
    status VARCHAR(20) DEFAULT 'booked' CHECK (status IN ('booked', 'confirmed', 'cancelled'))
);
```

8) FlightBooking

```
CREATE TABLE FlightBooking (
    flight_booking_id SERIAL PRIMARY KEY,
    user_id INT REFERENCES Users(user_id) ON DELETE CASCADE,
    flight_id INT REFERENCES Flight(flight_id) ON DELETE CASCADE,
    seat_number VARCHAR(10),
    status VARCHAR(20) DEFAULT 'booked' CHECK (status IN ('booked', 'confirmed', 'cancelled'))
);
```

9) Payment

```sql
CREATE TABLE Payment (
    payment_id SERIAL PRIMARY KEY,
    amount INT NOT NULL,
    payment_method VARCHAR(50),
    payment_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    hotel_booking_id INT REFERENCES HotelBooking(hotel_booking_id) ON DELETE CASCADE,
    flight_booking_id INT REFERENCES FlightBooking(flight_booking_id) ON DELETE CASCADE
);
```

10) Review

```sql
CREATE TABLE Review (
    review_id SERIAL PRIMARY KEY,
    user_id INT REFERENCES Users(user_id) ON DELETE CASCADE,
    hotel_id INT REFERENCES Hotel(hotel_id) ON DELETE CASCADE,
    rating INT CHECK (rating BETWEEN 1 AND 5),
    comment TEXT,
    review_date DATE DEFAULT CURRENT_DATE
);
```

TRIGGERS

```
CREATE OR REPLACE FUNCTION auto_calculate_booking_cost()
RETURNS TRIGGER AS $$
BEGIN
   NEW.total_cost := (NEW.check_out - NEW.check_in) * (
      SELECT rt.price_per_night FROM Room r
      JOIN RoomType rt ON r.room_type_id = rt.room_type_id
      WHERE r.room_id = NEW.room_id
   );
   IF NEW.total_cost <= 0 THEN
      NEW.total_cost := (SELECT rt.price_per_night FROM Room r JOIN
RoomType rt ON r.room_type_id = rt.room_type_id WHERE r.room_id =
NEW.room_id);
   END IF;
   RETURN NEW;
END;
$$ LANGUAGE plpgsql;

DROP TRIGGER IF EXISTS trg_auto_cost ON HotelBooking;

CREATE TRIGGER trg_auto_cost BEFORE INSERT OR UPDATE OF
check_in, check_out, room_id ON HotelBooking FOR EACH ROW
EXECUTE FUNCTION auto_calculate_booking_cost();
```

```
CREATE OR REPLACE FUNCTION handle_payments()
RETURNS TRIGGER AS $$
DECLARE v_cost INT; v_paid INT;
BEGIN
   IF NEW.hotel_booking_id IS NOT NULL THEN
      SELECT total_cost INTO v_cost FROM HotelBooking WHERE
hotel_booking_id = NEW.hotel_booking_id;
      SELECT SUM(amount) INTO v_paid FROM Payment WHERE
hotel_booking_id = NEW.hotel_booking_id;
      IF v_paid >= v_cost THEN UPDATE HotelBooking SET status =
'confirmed' WHERE hotel_booking_id = NEW.hotel_booking_id; END IF;
   ELSIF NEW.flight_booking_id IS NOT NULL THEN
      SELECT f.flight_price INTO v_cost FROM Flight f JOIN FlightBooking
fb ON f.flight_id = fb.flight_id WHERE fb.flight_booking_id =
NEW.flight_booking_id;
      SELECT SUM(amount) INTO v_paid FROM Payment WHERE
flight_booking_id = NEW.flight_booking_id;
      IF v_paid >= v_cost THEN UPDATE FlightBooking SET status =
'confirmed' WHERE flight_booking_id = NEW.flight_booking_id; END IF;
   END IF;
   RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_pay AFTER INSERT ON Payment FOR EACH
ROW EXECUTE FUNCTION handle_payments();
```

BALANCE REPORT

```
CREATE OR REPLACE VIEW BookingBalances AS
-- Part 1: Hotel Balances
SELECT
   'Hotel' AS Type,
   hb.hotel_booking_id AS ID,
   u.user_name,
   hb.total_cost AS Cost,
   COALESCE(SUM(p.amount), 0) AS Paid,
   (hb.total_cost - COALESCE(SUM(p.amount), 0)) AS Balance,
   hb.status
FROM HotelBooking hb
JOIN Users u ON hb.user_id = u.user_id
LEFT JOIN Payment p ON hb.hotel_booking_id = p.hotel_booking_id
GROUP BY hb.hotel_booking_id, u.user_name, hb.total_cost, hb.status
UNION ALL
-- Part 2: Flight Balances
SELECT
   'Flight' AS Type,
   fb.flight_booking_id AS ID,
   u.user_name,
   f.flight_price AS Cost,
   COALESCE(SUM(p.amount), 0) AS Paid,
   (f.flight_price - COALESCE(SUM(p.amount), 0)) AS Balance,
   fb.status
FROM FlightBooking fb
JOIN Users u ON fb.user_id = u.user_id
JOIN Flight f ON fb.flight_id = f.flight_id
LEFT JOIN Payment p ON fb.flight_booking_id = p.flight_booking_id
GROUP BY fb.flight_booking_id, u.user_name, f.flight_price, fb.status;

SELECT * FROM BookingBalances;
```

	type text	id integer	user_name character varying (100)	cost integer	paid bigint	balance bigint	status character varying (20)
1	Hotel	4	Maria Garcia	900	900	0	confirmed
2	Hotel	5	Admin Sam	700	0	700	booked
3	Hotel	1	Ahmed Ali	4000	6000	-2000	confirmed
4	Hotel	7	Hady	4000	4000	0	confirmed
5	Hotel	2	Sara Smith	800	100	700	booked
6	Hotel	3	Kenji Tanaka	100	100	0	confirmed
7	Flight	4	Maria Garcia	200	0	200	booked
8	Flight	3	Kenji Tanaka	350	0	350	booked
9	Flight	2	Sara Smith	1100	1100	0	confirmed
10	Flight	5	Admin Sam	400	0	400	booked
11	Flight	1	Ahmed Ali	800	800	0	confirmed

INSERTS

```
INSERT INTO Users (user_name, user_email, user_phone,
user_password, user_role) VALUES
('Ahmed Ali', 'ahmed@gmail.com', '01116622221', 'pass1', 'customer'),
('Sara Smith', 'sara@yahoo.com', '01116622222', 'pass2', 'customer'),
('Kenji Tanaka', 'kenji@outlook.jp', '01116622223', 'pass3', 'customer'),
('Maria Garcia', 'maria@gmail.com', '01116622224', 'pass4', 'customer'),
('Admin Sam', 'sam@travel.com', '01116622225', 'admin_pass', 'admin');

INSERT INTO Hotel (hotel_name, hotel_location, hotel_rating,
hotel_description) VALUES
('Burj Al Arab', 'Dubai', 5, 'Luxury'),
('The Ritz', 'London', 5, 'Classic'),
('Ibis Styles', 'Dubai', 3, 'Budget'),
('Park Hyatt', 'Tokyo', 5, 'Views'),
('Hotel Gran Via', 'Madrid', 4, 'Central');

INSERT INTO RoomType (hotel_id, room_type_name, price_per_night,
total_capacity) VALUES
(1, 'Royal Suite', 2000, 5),
(2, 'Superior', 400, 10),
(3, 'Economy', 100, 2),
(4, 'Zen Suite', 900, 4),
(5, 'Family', 350, 6);

INSERT INTO Room (room_type_id, room_number) VALUES
(1, '1001'),
(2, '201'),
(3, '305'),
(4, '701'),
(5, '501');
```

```
INSERT INTO Airline (airline_name) VALUES
('Emirates'),
('British Airways'),
('Japan Airlines'),
('Qatar Airways'),
('EgyptAir');

INSERT INTO Flight (airline_id, departure_city, arrival_city, departure_time,
arrival_time, flight_price, available_seats) VALUES
(1, 'Dubai', 'London', '2026-06-01 08:00', '2026-06-01 13:00', 800, 100),
(3, 'Tokyo', 'Dubai', '2026-06-15 09:00', '2026-06-15 15:00', 1100, 50),
(5, 'Cairo', 'Dubai', '2026-06-20 02:00', '2026-06-20 06:00', 350, 80),
(2, 'London', 'Paris', '2026-07-01 10:00', '2026-07-01 11:30', 200, 40),
(4, 'Dubai', 'Cairo', '2026-08-01 12:00', '2026-08-01 16:00', 400, 90);

INSERT INTO HotelBooking (user_id, room_id, check_in, check_out,
total_cost, status) VALUES
(1, 1, '2026-06-01', '2026-06-03', 0, 'booked'),
(2, 2, '2026-06-10', '2026-06-12', 0, 'booked'),
(3, 3, '2026-06-05', '2026-06-06', 0, 'booked'),
(4, 4, '2026-07-15', '2026-07-16', 0, 'booked'),
(5, 5, '2026-08-01', '2026-08-03', 0, 'booked');

INSERT INTO FlightBooking (user_id, flight_id, seat_number, status)
VALUES
(1, 1, '12A', 'booked'),
(2, 2, '05C', 'booked'),
(3, 3, '10F', 'booked'),
(4, 4, '01B', 'booked'),
(5, 5, '14D', 'booked');
```

```
INSERT INTO Payment (amount, payment_method, hotel_booking_id)
VALUES
(2000, 'Cash', 1), (2000, 'Credit Card', 1),
(100, 'PayPal', 2),
(100, 'Debit Card', 3),
(450, 'Cash', 4), (450, 'Credit Card', 4);

INSERT INTO Payment (amount, payment_method, flight_booking_id)
VALUES
(800, 'Credit Card', 1), (1100, 'PayPal', 2);

INSERT INTO Review (user_id, hotel_id, rating, comment) VALUES
(1, 1, 5, 'Perfect'),
(2, 2, 4, 'Good'),
(4, 4, 3, 'Average'),
(3, 3, 5, 'Cheap!'),
(5, 5, 2, 'Old');
```

CUSTOMER QUERIES

```
-- 7.1.1 Hotels and Rooms List
SELECT h.hotel_name, h.hotel_location, rt.room_type_name,
rt.price_per_night, rt.total_capacity as availible_rooms
FROM Hotel h
JOIN RoomType rt ON h.hotel_id = rt.hotel_id;
```

	hotel_name character varying (100)	hotel_location character varying (100)	room_type_name character varying (50)	price_per_night integer	availible_rooms integer
1	The Ritz	London	Superior	400	10
2	Ibis Styles	Dubai	Economy	100	2
3	Park Hyatt	Tokyo	Zen Suite	900	4
4	Hotel Gran Via	Madrid	Family	350	6
5	Burj Al Arab	Dubai	Royal Suite	2000	7

```
-- 7.1.2 Search for Hotels By location
SELECT hotel_name, hotel_rating, hotel_description
FROM Hotel
WHERE hotel_location ILIKE '%Dubai%';
```

	hotel_name character varying (100)	hotel_rating integer	hotel_description text
1	Burj Al Arab	5	Luxury
2	Ibis Styles	3	Budget

```
-- 7.1.3 Filter by Specific Room Type
SELECT h.hotel_name, rt.price_per_night, rt.total_capacity
FROM RoomType rt
JOIN Hotel h ON rt.hotel_id = h.hotel_id
WHERE rt.room_type_name = 'Royal Suite';
```

	hotel_name character varying (100)	price_per_night integer	total_capacity integer
1	Burj Al Arab	2000	7

```
-- 7.1.4 Cheapest Hotel in a Specific City
SELECT h.hotel_name, h.hotel_location, MIN(rt.price_per_night) as
starting_price
FROM Hotel h
JOIN RoomType rt ON h.hotel_id = rt.hotel_id
WHERE h.hotel_location = 'Dubai'
GROUP BY h.hotel_name, h.hotel_location
ORDER BY starting_price ASC
LIMIT 3;
```

	hotel_name character varying (100)	hotel_location character varying (100)	starting_price integer
1	Ibis Styles	Dubai	100
2	Burj Al Arab	Dubai	2000

```
-- 7.2.1 Find Available Flights by Destination
SELECT a.airline_name, f.departure_time, f.flight_price, f.available_seats
FROM Flight f
JOIN Airline a ON f.airline_id = a.airline_id
WHERE f.arrival_city = 'London' AND f.available_seats > 0;
```

	airline_name character varying (100)	departure_time timestamp without time zone	flight_price integer	available_seats integer
1	Emirates	2026-06-01 08:00:00	800	100

```
-- 7.2.2 Flights Cheaper than $500
SELECT departure_city, arrival_city, flight_price
FROM Flight
WHERE flight_price < 500
ORDER BY flight_price ASC;
```

	departure_city character varying (100)	arrival_city character varying (100)	flight_price integer
1	London	Paris	200
2	Cairo	Dubai	350
3	Dubai	Cairo	400

```
-- 7.2.3 Search for flight by date range
SELECT
    f.flight_id,
    a.airline_name,
    f.departure_city,
    f.arrival_city,
    f.departure_time,
    f.flight_price
FROM Flight f
JOIN Airline a ON f.airline_id = a.airline_id
WHERE f.departure_time BETWEEN '2026-06-01' AND '2026-06-30'
ORDER BY f.departure_time ASC;
```

	flight_id integer	airline_name character varying (100)	departure_city character varying (100)	arrival_city character varying (100)	departure_time timestamp without time zone	flight_price integer
1	1	Emirates	Dubai	London	2026-06-01 08:00:00	800
2	2	Japan Airlines	Tokyo	Dubai	2026-06-15 09:00:00	1100
3	3	EgyptAir	Cairo	Dubai	2026-06-20 02:00:00	350

```
-- Customer's Review History
SELECT h.hotel_name,u.user_name, r.rating, r.comment, r.review_date
FROM Review r
JOIN Hotel h ON r.hotel_id = h.hotel_id
JOIN Users u ON r.user_id = u.user_id
```

	hotel_name character varying (100)	user_name character varying (100)	rating integer	comment text	review_date date
1	Burj Al Arab	Ahmed Ali	5	Perfect	2026-03-25
2	The Ritz	Sara Smith	4	Good	2026-03-25
3	Park Hyatt	Maria Garcia	3	Average	2026-03-25
4	Ibis Styles	Kenji Tanaka	5	Cheap!	2026-03-25
5	Hotel Gran Via	Admin Sam	2	Old	2026-03-25

ADMIN REPORTS

```
-- A.1. Total Revenue per Hotel
SELECT h.hotel_name, SUM(p.amount) as total_earned , h.hotel_location
FROM Hotel h
JOIN RoomType rt ON h.hotel_id = rt.hotel_id
JOIN Room r ON rt.room_type_id = r.room_type_id
JOIN HotelBooking hb ON r.room_id = hb.room_id
JOIN Payment p ON hb.hotel_booking_id = p.hotel_booking_id
GROUP BY h.hotel_name,hotel_location
ORDER BY total_earned DESC;
```

	hotel_name character varying (100)	total_earned bigint	hotel_location character varying (100)
1	Burj Al Arab	10000	Dubai
2	Park Hyatt	900	Tokyo
3	The Ritz	100	London
4	Ibis Styles	100	Dubai

```
-- A.2. Low Inventory Alert (Rooms)
SELECT hotel_name, room_type_name, total_capacity
FROM Hotel h JOIN RoomType rt ON h.hotel_id = rt.hotel_id
WHERE total_capacity < 10;
```

	hotel_name character varying (100)	room_type_name character varying (50)	total_capacity integer
1	Ibis Styles	Economy	2
2	Park Hyatt	Zen Suite	4
3	Hotel Gran Via	Family	6
4	Burj Al Arab	Royal Suite	7

```
-- A.3. Flights Report
SELECT
    f.flight_id,
    f.arrival_city,
    f.available_seats,
    (SELECT COUNT(*) FROM FlightBooking fb WHERE fb.flight_id =
f.flight_id) AS seats_sold
FROM Flight f;
```

	flight_id [PK] integer	arrival_city character varying (100)	available_seats integer	seats_sold bigint
1	2	Dubai	50	1
2	3	Dubai	80	1
3	4	Paris	40	1
4	5	Cairo	90	1
5	1	London	100	1

```
-- A.4. Most Popular Hotel
SELECT h.hotel_name, COUNT(hb.hotel_booking_id) as booking_count
FROM Hotel h
JOIN RoomType rt ON h.hotel_id = rt.hotel_id
JOIN Room r ON rt.room_type_id = r.room_type_id
JOIN HotelBooking hb ON r.room_id = hb.room_id
GROUP BY h.hotel_name
ORDER BY booking_count DESC;
```

	hotel_name character varying (100)	booking_count bigint
1	Burj Al Arab	2
2	Ibis Styles	1
3	Park Hyatt	1
4	Hotel Gran Via	1
5	The Ritz	1

```
-- A.5. Customers with highest spending
SELECT user_name, SUM(Paid) as total_owed
FROM BookingBalances
GROUP BY user_name
HAVING SUM(Paid) > 0
ORDER BY total_owed DESC;
```

	user_name character varying (100)	total_owed numeric
1	Ahmed Ali	6800
2	Hady	4000
3	Sara Smith	1200
4	Maria Garcia	900
5	Kenji Tanaka	100

```
-- A.6. Airline Performance Report
SELECT a.airline_name, SUM(f.flight_price) as revenue_generated
FROM Airline a
JOIN Flight f ON a.airline_id = f.airline_id
JOIN FlightBooking fb ON f.flight_id = fb.flight_id
JOIN Payment p ON fb.flight_booking_id = p.flight_booking_id
GROUP BY a.airline_name
ORDER BY revenue_generated DESC;
```

	airline_name character varying (100)	revenue_generated bigint
1	Japan Airlines	1100
2	Emirates	800
3	EgyptAir	700
4	Qatar Airways	400

```
-- A.7. Monthly Revenue
SELECT TO_CHAR(payment_date, 'YYYY-MM') AS month, SUM(amount)
AS monthly_income
FROM Payment
GROUP BY month
ORDER BY month DESC;
```

	month text	monthly_income bigint
1	2026-03	13000

```
-- A.8. Top Rated Hotels
SELECT h.hotel_name, round(AVG(r.rating),0) as avg_rating
FROM Hotel h
JOIN Review r ON h.hotel_id = r.hotel_id
GROUP BY h.hotel_name
HAVING AVG(r.rating) >= 4;
```

	hotel_name character varying (100)	avg_rating numeric
1	Ibis Styles	5
2	Burj Al Arab	5
3	The Ritz	4

```
-- A.9. Payment Report (Today)
SELECT payment_method, SUM(amount)
FROM Payment
WHERE payment_date::DATE = CURRENT_DATE
GROUP BY payment_method;
```

	payment_method character varying (50)	sum bigint
1	Cash	2450
2	Credit Card	9250
3	Debit Card	100
4	PayPal	1200

HOTEL OR FLIGHT BOOKING

```
--(add new customer If needed)
INSERT INTO Users (user_name, user_email, user_phone,
user_password, user_role) VALUES
('Hady', 'hady@gmail.com', '01116622228', 'pass8', 'customer');

Select user_id,user_name
From Users
Where user_password = 'pass8'

-- 1. Customer Book Hotel
BEGIN;
   INSERT INTO HotelBooking (user_id, room_id, check_in, check_out,
total_cost, status)
   SELECT 6, 1, '2026-06-05', '2026-06-07', 0, 'booked'
   FROM Room WHERE room_id = 1 AND is_available = TRUE;

   UPDATE Room SET is_available = FALSE WHERE room_id = 1;
   UPDATE RoomType SET total_capacity = total_capacity - 1 WHERE
room_type_id = 1;
COMMIT;

-- 2. Customer Book Flight
BEGIN;
   INSERT INTO FlightBooking (user_id, flight_id, seat_number, status)
   SELECT 6, 1, '15B', 'booked'
   FROM Flight WHERE flight_id = 1 AND available_seats > 0;

   UPDATE Flight SET available_seats = available_seats - 1 WHERE
flight_id = 1;
COMMIT;
```

-- 3. Hotel Booking Details
SELECT ID as Booking_ID, user_name, Cost, Paid, Balance, status
FROM BookingBalances
WHERE Type = 'Hotel' AND user_name = 'Hady';

	booking_id integer	user_name character varying (100)	cost integer	paid bigint	balance bigint	status character varying (20)
1	7	Hady	4000	4000	0	confirmed

-- 4. Flight Booking Details
SELECT ID as Booking_ID, user_name, Cost, Paid, Balance, status
FROM BookingBalances
WHERE Type = 'Flight' AND user_name = 'Hady';

	booking_id integer	user_name character varying (100)	cost integer	paid bigint	balance bigint	status character varying (20)
1	7	Hady	800	0	800	booked

-- 5. Hotel Payment
```
BEGIN;
   INSERT INTO Payment (amount, payment_method, hotel_booking_id)
   VALUES (4000, 'Credit Card', (SELECT hotel_booking_id FROM
HotelBooking WHERE user_id = 6 ORDER BY 1 DESC LIMIT 1));
COMMIT;
```

-- 6. Flight Payment
```
BEGIN;
   INSERT INTO Payment (amount, payment_method, flight_booking_id)
   VALUES (400, 'Cash', (SELECT flight_booking_id FROM FlightBooking
WHERE user_id = 6 ORDER BY 1 DESC LIMIT 1));
COMMIT;
```

-- 7. Update Flight Seat
UPDATE FlightBooking SET seat_number = '01A'
WHERE user_id = 6 AND status = 'booked';

SELECT * FROM FlightBooking WHERE user_id = 6;

	flight_booking_id [PK] integer	user_id integer	flight_id integer	seat_number character varying (10)	status character varying (20)
1	7	6	1	15B	booked

-- 8. Update Hotel Dates
UPDATE HotelBooking SET check_out = '2026-06-08'
WHERE user_id = 6 AND status = 'booked';

SELECT * FROM HotelBooking WHERE user_id = 6;

	hotel_booking_id [PK] integer	user_id integer	room_id integer	check_in date	check_out date	total_cost integer	status character varying (20)
1	7	6	1	2026-06-05	2026-06-07	4000	confirmed

-- 9. Deleting Hotel Booking
BEGIN;
UPDATE Room SET is_available = TRUE WHERE room_id = (SELECT room_id FROM HotelBooking WHERE hotel_booking_id = 6);
UPDATE RoomType SET total_capacity = total_capacity + 1 WHERE room_type_id = 1;
DELETE FROM HotelBooking WHERE hotel_booking_id = 6 AND status = 'booked';
COMMIT;

-- 10. Deleteing Flight Booking
BEGIN;
UPDATE Flight SET available_seats = available_seats + 1 WHERE flight_id = 1;
DELETE FROM FlightBooking WHERE flight_booking_id = 6 AND status = 'booked';
COMMIT;

-- 11. Full Report

SELECT * FROM BookingBalances;

	type text	id integer	user_name character varying (100)	cost integer	paid bigint	balance bigint	status character varying (20)
1	Hotel	4	Maria Garcia	900	900	0	confirmed
2	Hotel	5	Admin Sam	700	0	700	booked
3	Hotel	1	Ahmed Ali	4000	6000	-2000	confirmed
4	Hotel	7	Hady	4000	4000	0	confirmed
5	Hotel	2	Sara Smith	800	100	700	booked
6	Hotel	3	Kenji Tanaka	100	100	0	confirmed
7	Flight	7	Hady	800	0	800	booked
8	Flight	4	Maria Garcia	200	0	200	booked
9	Flight	3	Kenji Tanaka	350	0	350	booked
10	Flight	2	Sara Smith	1100	1100	0	confirmed
11	Flight	5	Admin Sam	400	0	400	booked
12	Flight	1	Ahmed Ali	800	800	0	confirmed